LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Quick Revision: Sound

Key ideas

- Sound is produced by a **vibrating** object. In humans, sound comes from the vibrating vocal cords in the **larynx** (voice box).
- Sound needs a **medium** (solid, liquid or gas). It cannot travel through a vacuum.
- We hear when sound makes the **eardrum** vibrate; the vibrations are passed on to the inner ear and then to the brain.

Term	Meaning	Decides
Amplitude	Size of the vibration	Loudness, measured in decibels (dB)
Frequency	Number of vibrations per second, in hertz (Hz)	Pitch: higher frequency, shriller sound
Time period	Time for one vibration, in seconds	Time period = 1 ÷ frequency

Human hearing: about **20 Hz to 20,000 Hz**. Below 20 Hz is infrasonic; above 20,000 Hz is ultrasonic (dogs can hear some of it).
Sounds above about **80 dB** become physically painful.

Noise and noise pollution

- Unpleasant sounds are **noise**; pleasant sounds are music.
- Sources of noise pollution: vehicle horns, loudspeakers, firecrackers, machines, construction work.
- Effects: lack of sleep, high blood pressure, anxiety, and even loss of hearing.
- Control: silencers on vehicles and machines, keeping TVs and music low, planting trees along roads, keeping factories away from homes.

Quick check

1. A pendulum makes 40 oscillations in 4 seconds. Find its frequency and time period.

2. Why can't astronauts on the Moon talk to each other directly, without radios?